

Sean Luka Girgis

Senior AWS Data Quality / ETL Engineer

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Professional Summary

Senior AWS Data Engineer with 20+ years of enterprise IT experience across Python, SQL, PySpark/Spark, ETL/ELT pipelines, data quality, reconciliation, schema validation, telemetry analytics, and reporting platforms. Strong background designing and validating data workflows using Python, SQL, PySpark, Pandas, Oracle, AWS S3/Glue/Athena/Redshift patterns, data warehousing, operational monitoring, and pipeline reliability practices. Experienced partnering with engineering, BI, DevOps, and business teams to define quality checks, troubleshoot pipeline issues, validate transformations, document workflows, and improve trust in analytics-ready datasets.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Architected automated Python/Pandas ETL pipelines ingesting P95 telemetry from 6,000+ infrastructure endpoints, replacing manual processes with repeatable, validated data workflows.
- Designed data validation, reconciliation, and quality checks across monitoring feeds to improve reliability of enterprise capacity, forecasting, and executive reporting outputs.
- Built hybrid AWS/Oracle data platform patterns using S3 landing zones, AWS Glue ETL concepts, Redshift analytics workloads, and Oracle schemas for historical retention and reporting.
- Used PySpark/Spark patterns to process large-scale telemetry and support parallel forecasting workloads across high-volume infrastructure datasets.
- Integrated BMC TrueSight/TSCO, CA APM, and AppDynamics feeds into unified reporting datasets with transformation checks, source reconciliation, and data integrity controls.
- Designed and optimized SQL/Oracle reporting schemas to support data warehousing, capacity planning, trend analysis, dashboarding, and operational risk reporting.
- Developed Prophet and scikit-learn forecasting models to identify capacity bottleneck risk, using validated time-series inputs and documented assumptions.
- Created dashboards and reporting outputs using Streamlit, matplotlib, seaborn, and plotly to communicate data quality, performance, utilization, and capacity insights to stakeholders.
- Maintained documentation for data definitions, transformation logic, validation rules, pipeline behavior, and operational workflows to support governance and repeatability.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring for Brand.com and critical systems, improving visibility into production performance and operational telemetry quality.
- Led the FAST project, mining real-user and synthetic performance metrics to identify optimization opportunities and deliver actionable findings to engineering teams.
- Integrated HP Performance Center with Dynatrace to correlate load-test behavior, APM telemetry, and infrastructure metrics for performance validation.

- Implemented End-User Monitoring and transaction tracing across web and mobile systems to improve visibility into user journeys, API behavior, and application performance.
- Participated in AWS cloud migration readiness discussions by evaluating monitoring coverage, operational risks, and data requirements.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as CA APM subject matter expert for a large financial-services enterprise with 50+ Enterprise Managers and 4,000–6,000 instrumented agents.
- Designed dashboards, alerts, SLA thresholds, and management modules used to monitor enterprise application performance and support operational decision-making.
- Built Perl and Korn Shell data-extraction scripts to automate performance reporting and reduce manual weekly exports.
- Partnered with J2EE/WebLogic and infrastructure teams to diagnose bottlenecks using APM telemetry, SQL/Oracle data, thread behavior, and infrastructure KPIs.
- Delivered technical documentation, onboarding guidance, and client training for monitoring standards, deployment processes, and production troubleshooting.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed J2EE telecom web applications under load to identify throughput limits, response-time degradation, and resource bottlenecks including JDBC connections, threads, heap memory, CPU, and garbage collection.
- Used Java/J2EE, JMX monitoring, Wily Introscope, and thread-dump automation practices to provide real-time visibility during performance validation.
- Documented performance baselines, regression thresholds, findings, and recommendations used by engineering teams for release-gate decisions.
- Collaborated with application, testing, and infrastructure teams to troubleshoot performance defects and validate remediation outcomes.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining strict performance and reliability expectations.
- Built C++/OCCI/OCI validation tooling to verify migration correctness and performance-sensitive behavior across large transaction datasets.
- Optimized core transaction processing using C++, OCCI, OCI, SQL, and Oracle, reducing physical hardware footprint while preserving sub-second query latency.
- Automated schema conversion and data validation workflows to protect data integrity across relational database migration paths.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (AMDOCS projects) (2001 – 2007)

- Developed high-availability telecom billing interfaces using C, C++, Pro*C, PL/SQL, Oracle, XML, socket programming, POSIX threads, and Unix environments.
- Improved throughput and database performance in billing processes through SQL tuning, sequence optimization, memory reduction, and interface-level performance analysis.
- Built and maintained reporting, invoicing, settlement, and integration components across full billing lifecycle systems for large-scale telecom operations.

- Worked with enterprise integration and messaging patterns including MQ Series, XML-based workflows, WebLogic/WebSphere, and Unix automation.

Education

High Diploma / Post-Graduate Diploma in Computer Engineering Technology / Computer Science

Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering

Zagazig University, Egypt / Cairo, Egypt

Skills

Languages: Python, SQL, PySpark, PL/SQL, C++, C, Java, Perl, Korn Shell, XML, JSON

Flagship Projects

AWS Data Quality and Lakehouse Validation Patterns

Designed AWS-oriented data platform patterns using S3, Glue, Athena, Redshift, PySpark, Parquet, and validation checks for cloud-scale ingestion, transformation, analytics, and governed reporting.

Technologies: AWS S3, AWS Glue, Athena, Amazon Redshift, PySpark, Parquet, Data Quality

Performance Telemetry and Capacity Reporting Platform

Built Python and SQL-driven reporting workflows to turn monitoring and infrastructure telemetry into validated datasets, dashboards, executive reports, and trend analysis.

Technologies: Python, SQL, Oracle, BMC TrueSight, AppDynamics, CA APM, Streamlit

HorizonScale — AI Capacity Forecasting Engine

Designed a forecasting engine using Python, PySpark, Prophet, scikit-learn, and Streamlit to transform enterprise telemetry into capacity-risk predictions and operational decision support.

Technologies: Python, PySpark, Prophet, scikit-learn, Streamlit, Telemetry Data, Forecasting

AI-Powered Job Search Pipeline

Built an agentic Python pipeline for job intake, semantic duplicate detection, LLM scoring, JSON artifact generation, quality gates, and tracking using FAISS, sentence-transformers, Grok API, and Claude Code workflows.

Technologies: Python, FAISS, Sentence Transformers, LLM APIs, RAG, Agentic Workflows, JSON Contracts